Basic Principles of Medicine 2
Module : Digestion and Metabolism
Lecture No: 37

Title: Nutrition and intensive care

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Nutrition and intensive care

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Access Medicine: Schwartz's Principles of Surgery, 10e ;

Chapter 2. Systemic Response to Injury and Metabolic Support

Objectives: Nutrition and intensive care

SOM. MK.III.BPM2.3.DM.3.BCHM.1353	Generalize features of hypermetabolic response or Systemic inflammatory response syndrome (SIRS) and list its causes
SOM. MK.III.BPM2.3.DM.3.BCHM.1354	Explain the role of hormones (epinephrine, cortisol and insulin) and inflammatory mediators (cytokines and PGE2) in patients with critical illness.
SOM. MK.III.BPM2.3.DM.3.BCHM.1355	Correlate hormonal and metabolic changes (carbohydrate, protein and lipid) following trauma in the three phases: Ebb phase (Before resuscitation), Flow phase (Adrenergic-corticoid phase), Recovery phase (convalescent/ anabolic phase)
SOM. MK.III.BPM2.3.DM.3.BCHM.1356	Indicate significance of acute phase proteins in response to critical illness
SOM. MK.III.BPM2.3.DM.3.BCHM.1357	Explore significance of nutritional support and role of glutamine, arginine, antioxidants and omega-3 fatty acid supplementation (Immunomodulators and immunonutrition)
SOM. MK.III.BPM2.3.DM.3.BCHM.1358	Identify minerals (Copper and zinc) and vitamins (vitamin C) important for wound healing in post-operative patients
SOM. MK.III.BPM2.3.DM.3.BCHM.1359	Distinguish pros and cons of enteral vs parenteral nutrition
SOM. MK.III.BPM2.3.DM.3.BCHM.1360	Compare and contrast metabolic response to injury and starvation

Recommended Reading

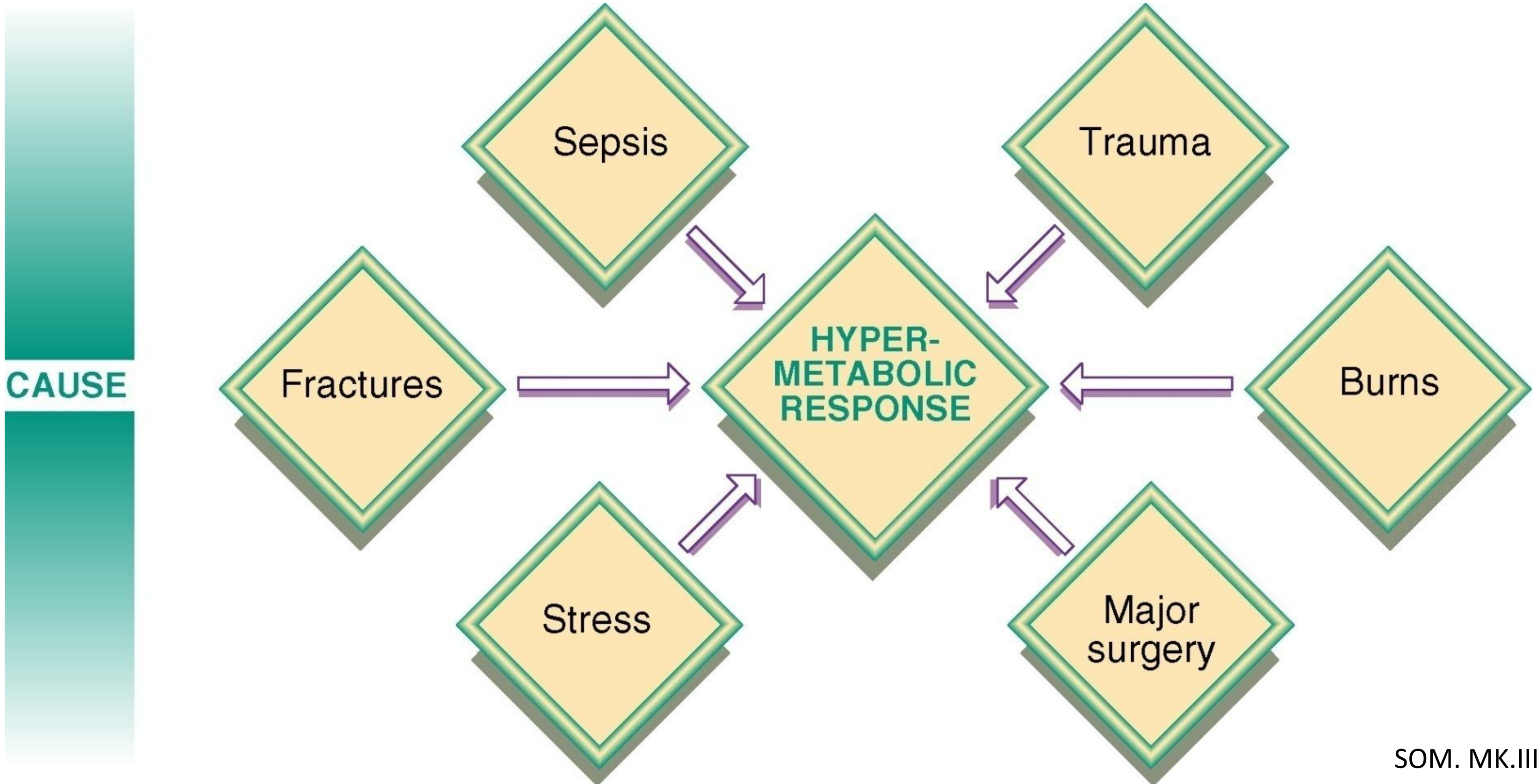
- Access Medicine: Schwartz's Principles of Surgery, 11e ; Chapter 2. Systemic Response to Injury and Metabolic Support [Systemic Response to Injury and Metabolic Support | Schwartz's Principles of Surgery, 11e | AccessMedicine | McGraw Hill Medical (oclc.org)]
- The sections in the Chapter that may be of interest:
  - CENTRAL NERVOUS SYSTEM REGULATION OF INFLAMMATION IN RESPONSE TO INJURY
  - MEDIATORS OF INFLAMMATION: Eicosanoids
  - SURGICAL METABOLISM
  - NUTRITION IN THE SURGICAL PATIENT
  - Enteral Nutrition
  - Parenteral nutrition

Mr. Katta Bolic (modified from Mark's)

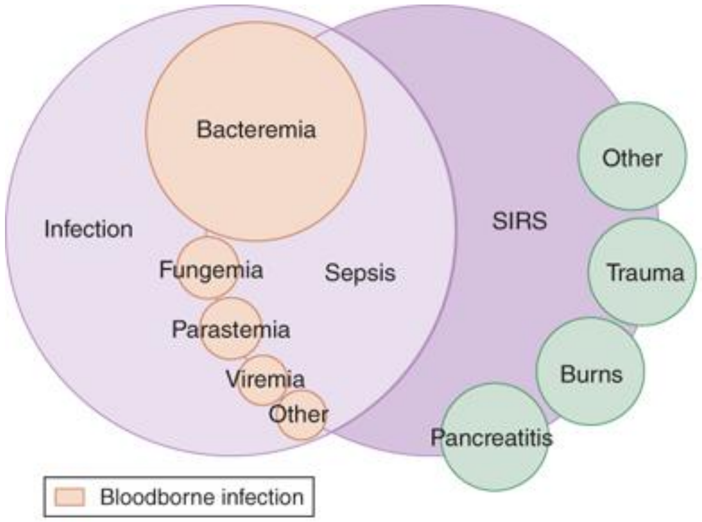
- **Katta Bolic**, a 62-year-old homeless man, brought to emergency department
- **Fever** 103°F, severely dehydrated, Marked **muscle wasting**
- Heart rate very rapid, **blood pressure is low** (85/46 mm Hg)
- Abdomen distended and NO bowel sounds indicating **peritonitis**
- pH: 7.3 (7.4); PCO_2: 25 (40); HCO3-: 17 (24); Anion gap: 29 **(Increased Lactate)**
- Intravenous glucose, saline, and parenteral broad-spectrum antibiotics are begun. Radiographs suggested a bowel perforation.
- Blood cultures grew *Escherichia coli* (**septicemia**)
- Team (surgeons, internists, and nutritionists) develop a therapeutic plan
- **Negative nitrogen balance**
- **Degradation of muscle protein** and release of amino acids into blood
- **Serum C-reactive protein** levels elevated (acute phase response)
- Weight loss: **Loss of appetite**, **Hypermetabolism** and **muscle proteolysis**

Systemic Inflammatory Response Syndrome (SIRS)

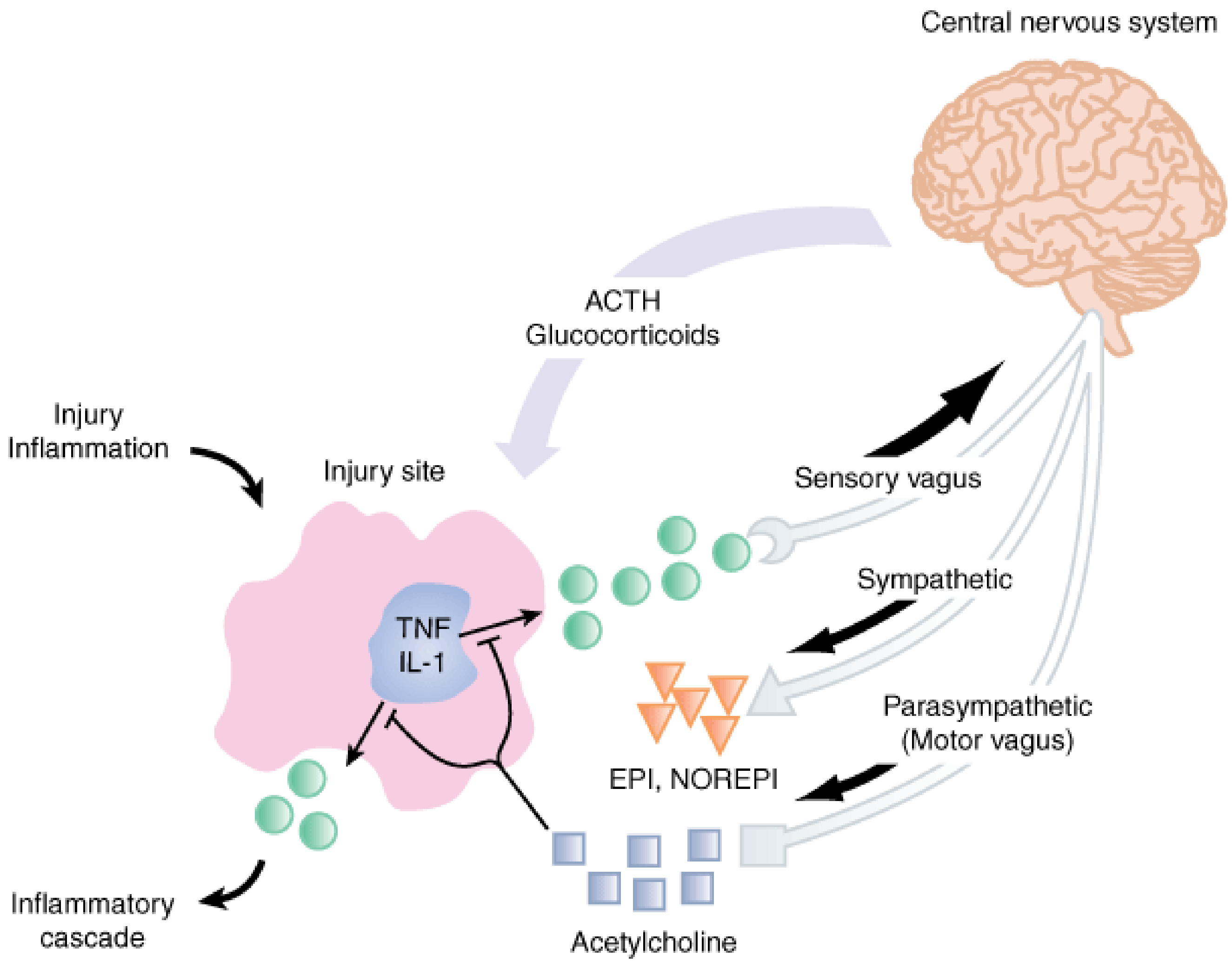
- Clinical response to nonspecific insult of infectious or noninfectious origin (Hypermetabolic response)
- Similar physiologic and metabolic changes, regardless of cause



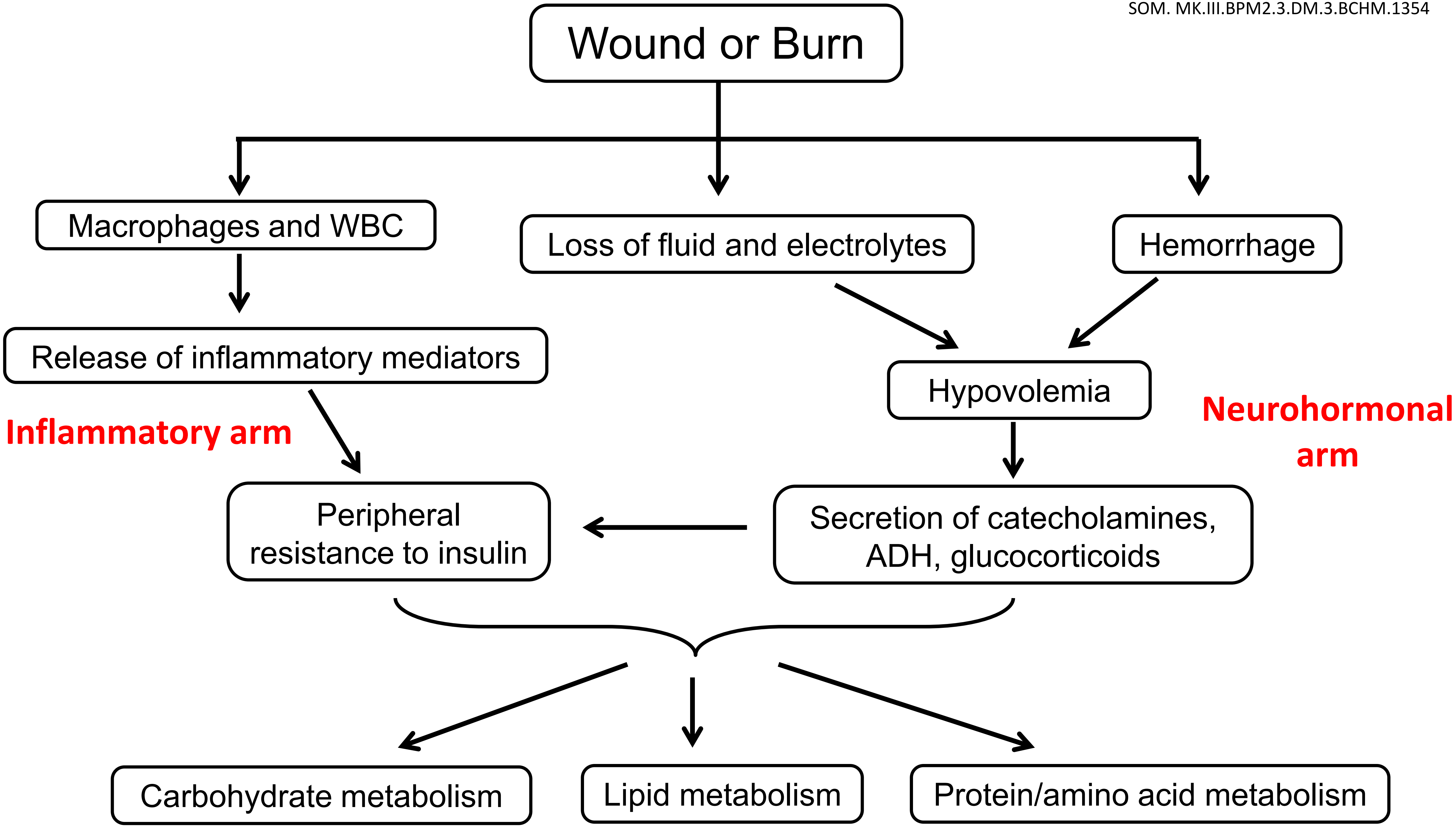
Systemic Inflammatory Response Syndrome (SIRS)



Source: J.E. Tintinalli, J.S. Stapczynski, O.J. Ma, D.M. Yealy, G.D. Meckler, D.M. Cline: Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th Edition
www.accessmedicine.com



Source: Brunicardi FC, Andersen DK, Billiar TR, Dunn DL, Hunter JG, Matthews JB, Pollock RE: *Schwartz's Principles of Surgery, 9th Edition*: http://www.accessmedicine.com


Wound or Burn
Macrophages and WBC
Loss of fluid and electrolytes
Hemorrhage
Release of inflammatory mediators
Hypovolemia
Inflammatory arm
Neurohormonal arm
Peripheral resistance to insulin
Secretion of catecholamines, ADH, glucocorticoids
Carbohydrate metabolism
Lipid metabolism
Protein/amino acid metabolism

Mediators of stress response

- Neurohormonal (Neuroendocrine) arm
  - Catecholamines, ACTH, glucocorticoids, glucagon, ADH, aldosterone
  - Sympathetic nervous system active and hypothalamic-pituitary axis
- Inflammatory arm (by macrophages/ phagocytes)
  - Release of Cytokines, eicosanoids (PGE_2)
  - Increased oxidative stress and formation of reactive oxygen species

Hormonal mediators of the stress response

- Aldosterone - Renal sodium reabsorption
- Antidiuretic hormone (ADH) - Renal tubular water reabsorption
- ACTH - Releases cortisol (glucocorticoids) (stimulates lipolysis, mobilizes amino acids from skeletal muscles and stimulates gluconeogenesis)
- Catecholamines - Epinephrine and norepinephrine from adrenal medulla stimulate hepatic glycogenolysis, fat mobilization, gluconeogenesis

Role of inflammatory mediators (cytokines)

- Inflammatory mediators: Interleukins, tumor necrosis factor (TNF), eicosanoids (PGE_2) released by activated phagocytes/ macrophages due to tissue damage, infection, inflammation
- **Local** (**paracrine)** effects
  - Promote wound healing by ingrowth of fibroblasts
  - Stimulate angiogenesis
  - Increase white cell counts and facilitate white cell migration
  - Localize wound
- **Systemic** effects
  - Mobilize amino acids, stimulate acute phase protein synthesis by liver
  - Responsible for fever (increases metabolic rate)
  - Responsible for pain
  - Contribute to insulin resistance

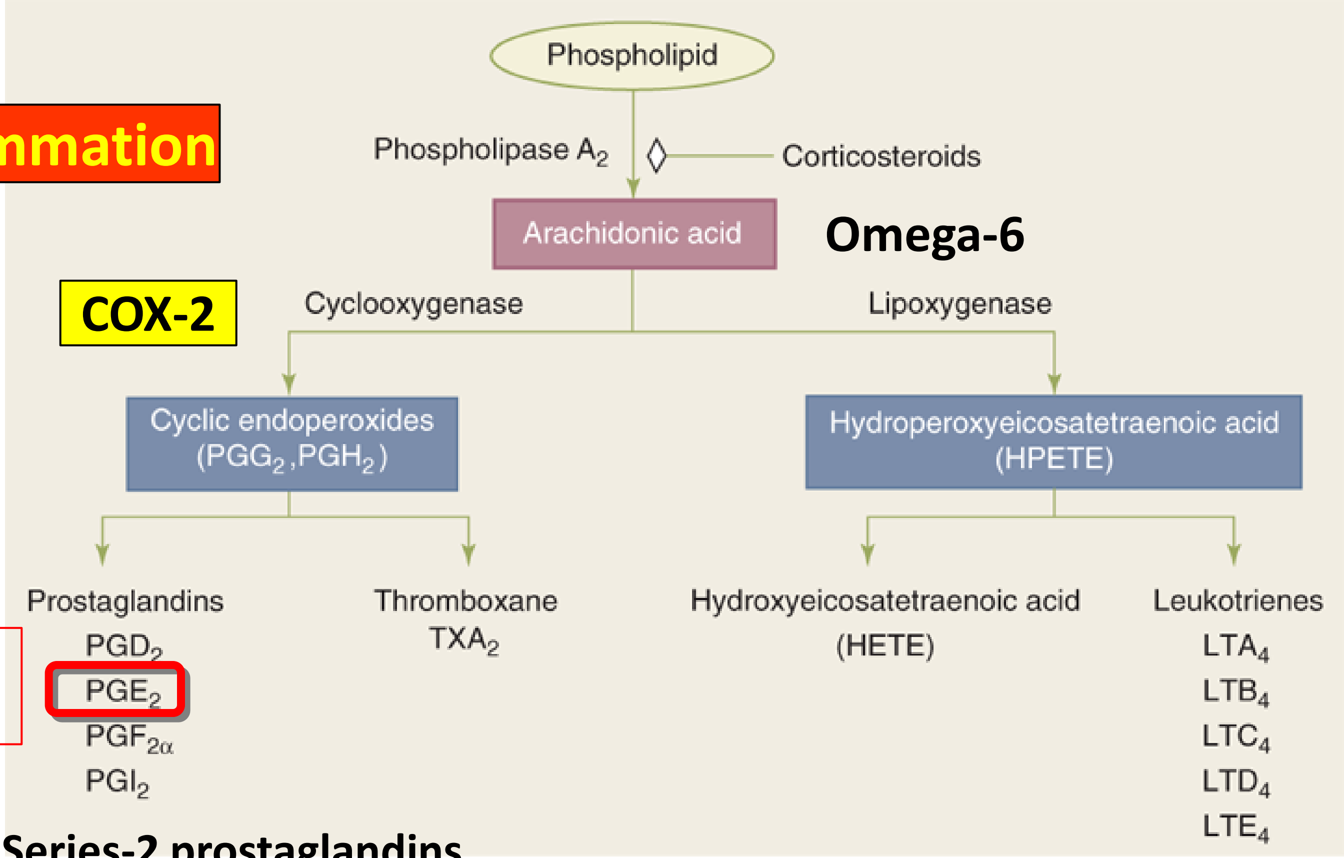
Eicosanoids

NSAID – reduces pain, fever and inflammation

Inflammation

COX-2

Omega-6



Mediators of inflammation

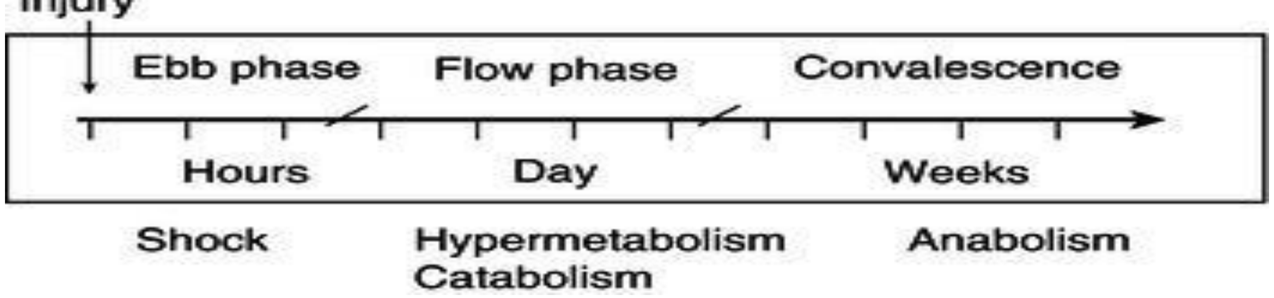
Series-2 prostaglandins

A

Source: F.C. Brunicardi, D.K. Andersen, T.R. Billiar, D.L. Dunn, L.S. Kao, J.G. Hunter, J.B. Matthews, R.E. Pollock: Schwartz's Principles of Surgery, 11e

Immunomodulators: Omega-3 fatty acid supplements produce series-3 prostaglandins (anti-inflammatory)

Phases following critical injury



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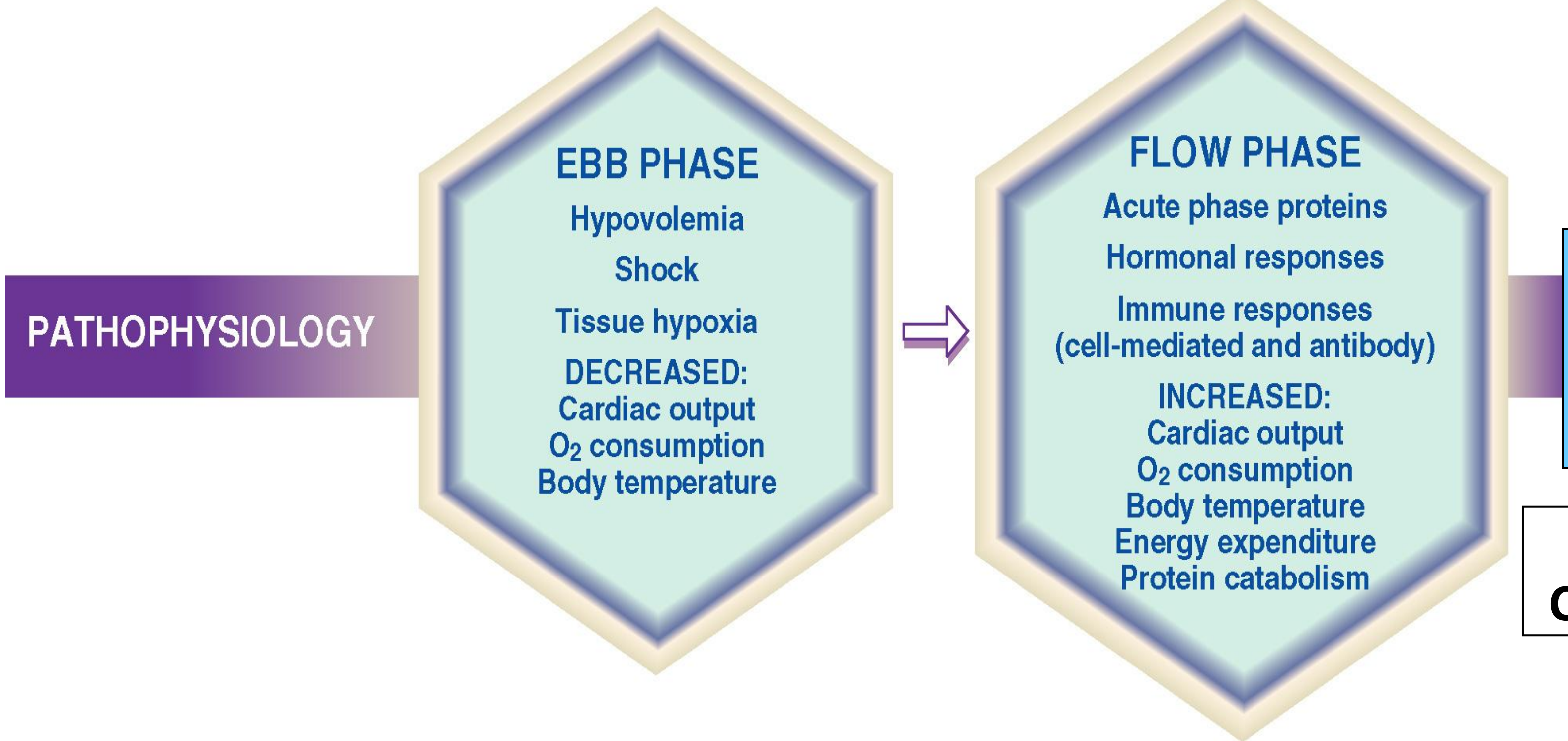


Anabolic phase

Recovery/ Convalescent phase

Ebb phase

Before resuscitation

Flow phase

Adrenergic - corticoid

Ebb Phase

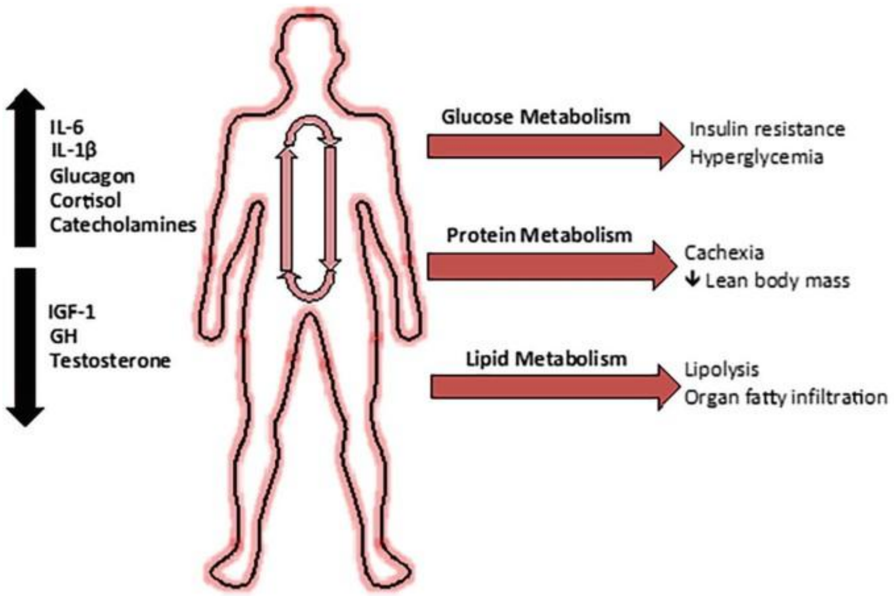
- Few hours after injury (**transient**)
- Hypovolemia, shock, tissue hypoxia
- Decreased cardiac output (lactic acidosis due to low tissue oxygen)
  - Anaerobic glycolysis forms lactate
  - Impaired circulation can disrupt Cori cycle (glucose-lactate cycle)
  - Lactic acidosis causes high anion gap metabolic acidosis
    - Low pH, HCO_3^- low; PCO_2 low; Increased anion gap
- Decreased oxygen consumption (decreased metabolic rate)
- Lower body temperature
- Glucagon, epinephrine and cortisol elevated; **Insulin levels are low**
- Hyperglycemia (proportional to severity of injury/ stress)
- **Low insulin levels** with slightly increased glucose production

Flow Phase (Adrenergic – Corticoid)

- Lasts up to 2 weeks following injury (Longer in severe injury)
- Increased cardiac output; Increased pulse rate
- Increased body temperature (fever)
- Increased energy expenditure (Increased metabolic rate/ **Hypermetabolism)**
- Increase in catecholamines, glucagon, cortisol and cytokines: Increased counter regulatory hormones →Insulin receptor substrates (IRS-1) modified → **INSULIN resistance**
- Increased insulin secretion

Flow Phase (Adrenergic – Corticoid)

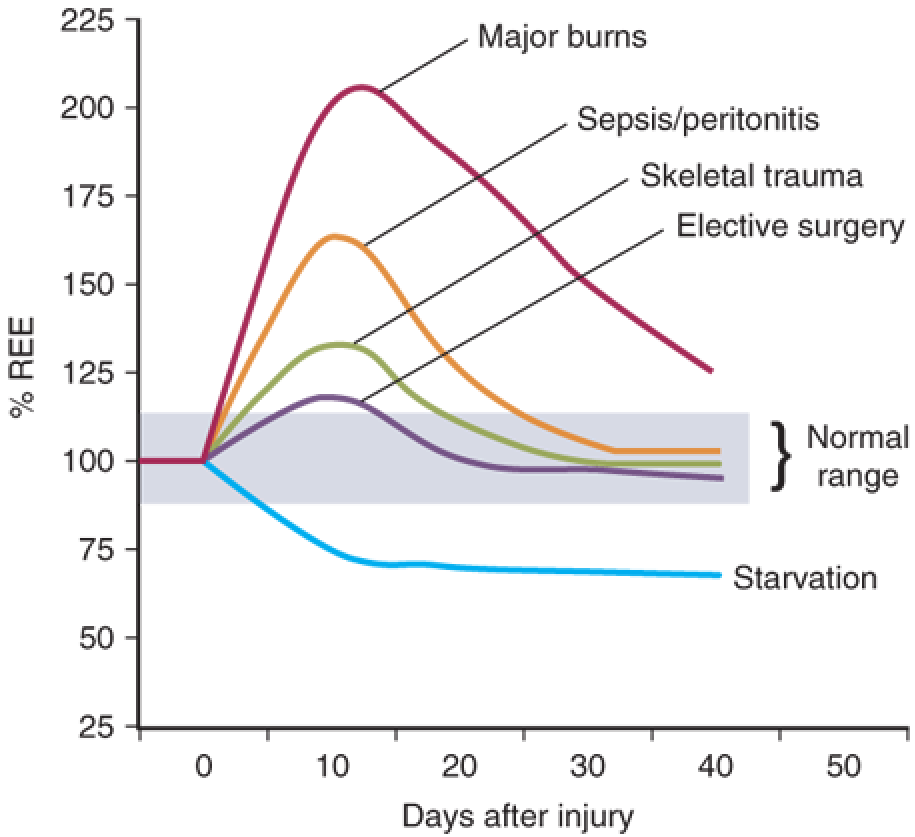
- Insulin resistance; Increased glucose production and hyperglycemia (Carbohydrate metabolism)
- Protein catabolism (Protein metabolism)
- Increased adipose tissue lipolysis (Lipid metabolism)
- Mobilize energy stores → for wound healing and recovery (Coordinated metabolic response)
- **Reprioritize** resources (muscle, adipose tissue) for key organs/ functions (Liver, immune system and WOUND)



SOM. MK.III.BPM2.3.DM.3.BCHM.1355

Metabolic rate in 'Flow phase'

- Marked increased metabolic rate (hypermetabolism)
- Metabolic rate: CO_2 produced/min OR O_2 consumed/min
- Increased oxygen utilization
- More severe injury → Higher metabolic rate
- In **prolonged starvation,** adaptive decrease in metabolic rate (adaptation increases survival)



Source: F.C. Brunicardi, D.K. Andersen, T.R. Billiar, D.L. Dunn, L.S. Kao, J.G. Hunter, J.B. Matthews, R.E. Pollock: Schwartz's Principles of Surgery, 11e

Flow phase: Caloric Requirements

Disease process	kcal/day
Basal	1,450
Post-op. (uncomplicated)	1,500–1,700
Sepsis	2,000–2,400
Multiple trauma (ventilator)	2,200–2,600
Major burn	2,500–3,000

Energy needs increase as severity of illness increases

Approximate caloric requirement in flow phase: 30-35 Cals/ kg/day

Avoid oversupplementation

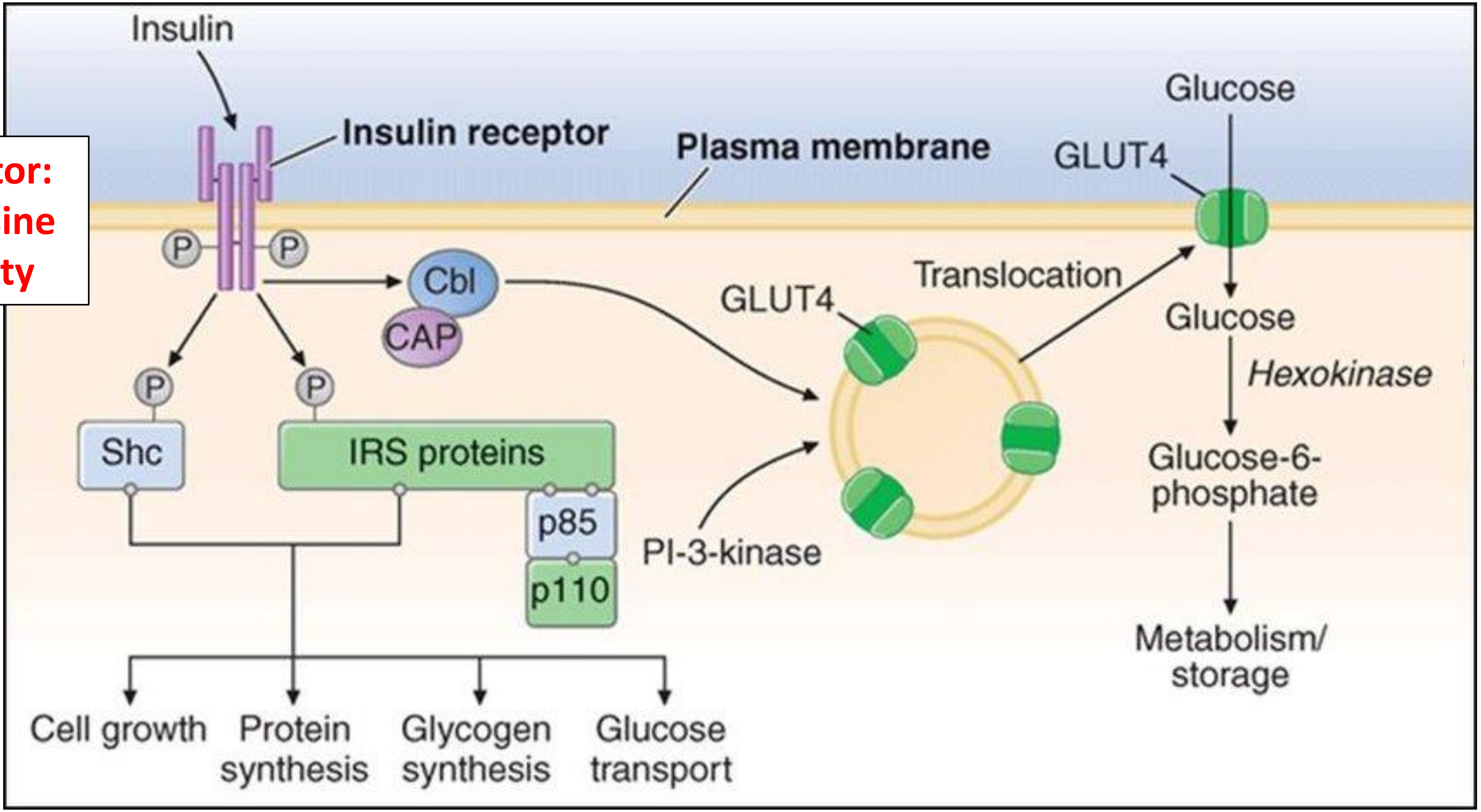
Carbohydrates and fat supplements provide energy

Changes in carbohydrate metabolism (Flow phase)

- **Hyperglycemia** (glucagon, epinephrine and glucocorticoids)
- Target tissues (adipose tissue, muscle and liver) show 'insulin resistance' → Increased insulin secretion
  - Elevated **epinephrine, cortisol and cytokines** (counter regulatory hormones) – Modifies **intracellular IRS-1**
- Insulin resistance (Insulin less effective)
  - Increases hepatic gluconeogenesis from amino acids from muscle proteolysis (epinephrine and glucocorticoids)
  - Reduces glucose uptake by muscle and adipose tissue (fewer GLUT-4)

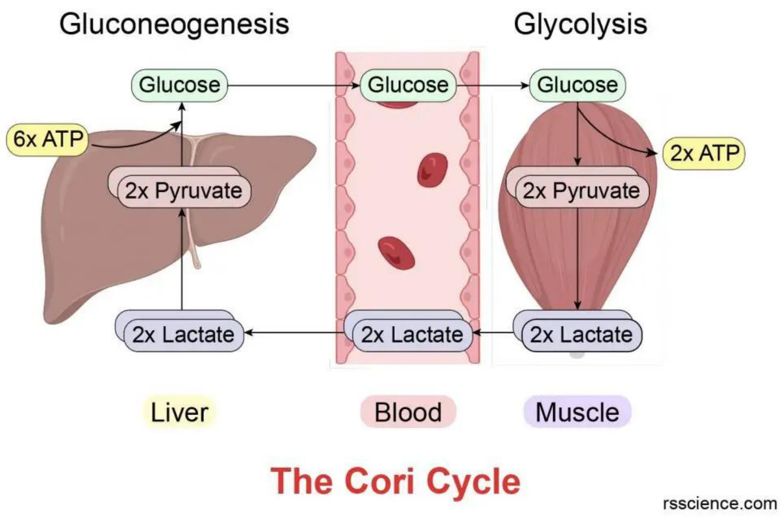
Mechanism of insulin action on target cells

Insulin receptor: Intrinsic tyrosine kinase activity



Changes in carbohydrate metabolism - Lactic acidosis

- Glucose used by injured tissue/ wound
  - Wound and injured tissue anaerobic glycolysis
  - Lactate recycled to liver by Cori cycle
- Lactic acidosis in critically injured:
  - Poor tissue oxygenation resulting in anaerobic glycolysis
  - Poor blood flow impairs Cori cycle
  - Lactic acidosis: indicator of poor prognosis
  - Monitor serum lactate as prognostic indicator
  - Lactic acidosis: **High anion gap metabolic acidosis**
    - Low pH; HCO_3^- -Low (why?); PCO_2-Low (compensation); Anion gap increased



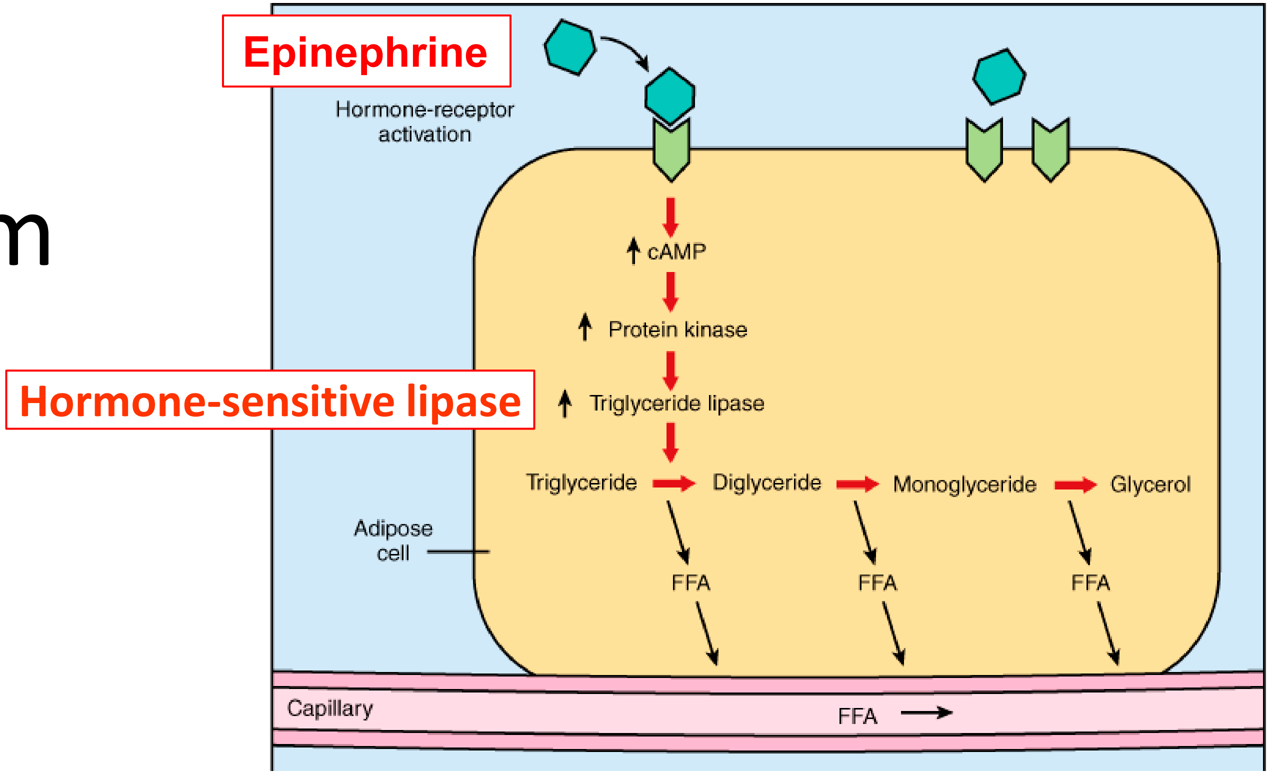
Carbohydrate metabolism – Diabetic patients in flow phase

- Plasma glucose homeostasis in diabetics difficult to achieve due to additional **insulin resistance**
- Type 2 diabetics may present in **hyperosmolar hyperglycemic state or ketosis**
- Type 1 diabetics may have **ketoacidosis** following injury/ infection
  - Do not respond to normal doses of insulin due to insulin resistance
- In diabetes mellitus (types 1 and 2), there is **worsening of diabetes** following injury (additional insulin resistance)
- Require insulin injections to manage hyperglycemia during infections or surgery

Comparison of Carbohydrate metabolism

Critical illness (Flow phase)	Prolonged starvation
• Blood glucose level: Increased	• Blood glucose level: Decreased
• Insulin levels: High and insulin resistance	• Insulin levels: Low
• Energy for the Brain: Glucose	• Energy for the Brain: Glucose + Ketone bodies
• Energy for the Wound: Glucose	
• Glucose uptake by adipose and muscle (GLUT4): Low due to insulin resistance	• Glucose uptake by adipose and muscle (GLUT4): Low due to low levels of insulin
• Gluconeogenesis: Highly active to maintain the high blood glucose levels	• Gluconeogenesis: Active to maintain the blood glucose levels

Changes in lipid metabolism



- Triglycerides mobilized and oxidized at higher rate
  - Epinephrine/ norepinephrine, cortisol activate hormone-sensitive lipase (Phosphorylated state)
  - **Adipose tissue lipolysis active**
  - Plasma free fatty acid levels: Elevated

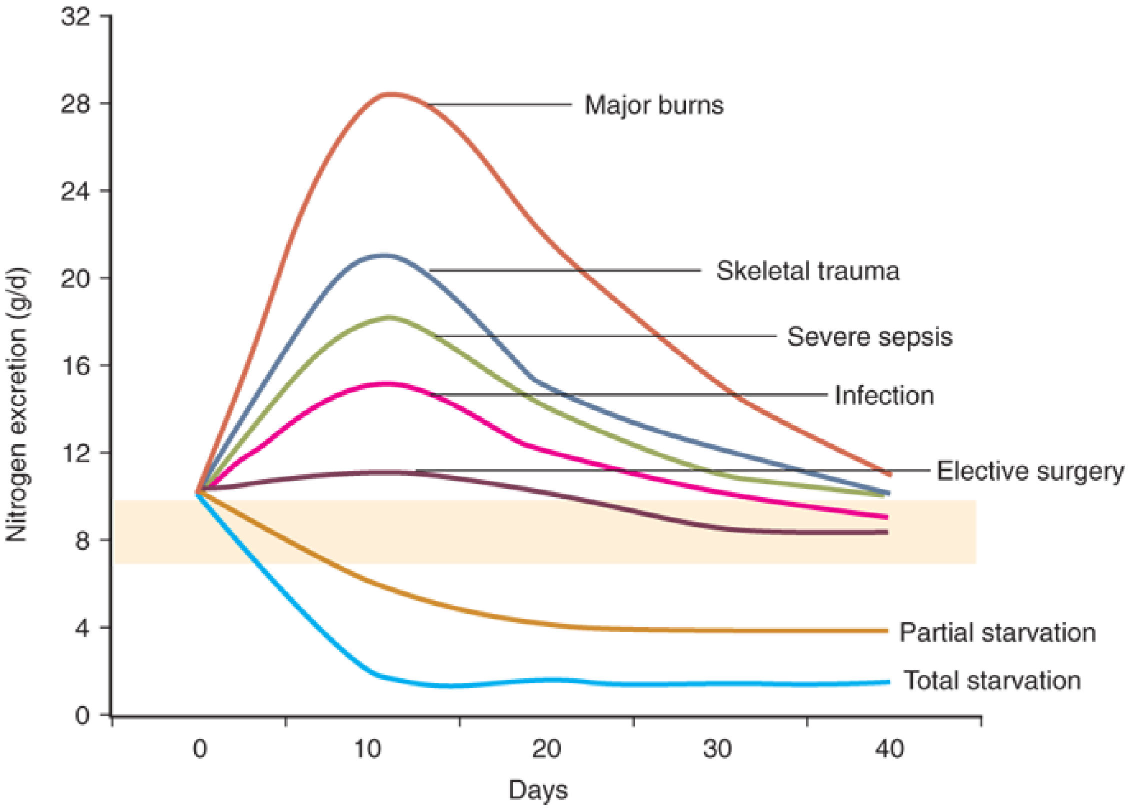
Changes in lipid metabolism

- Plasma free fatty acid levels elevated: Energy for skeletal and cardiac muscle
- Ketosis is not observed (blunted)
  - High levels of insulin; Insulin inhibits ketogenesis
  - Peripheral tissues use ketone bodies (due to higher metabolic rate)
  - Ketogenesis is inversely proportional to severity of injury
- Extent of lipolysis not proportionate to plasma ketone body levels **(Blunted ketogenesis)**

Comparison of Lipid metabolism

Critical illness (Flow phase)	Prolonged starvation
• Hormone sensitive lipase: Phosphorylated and very active due to epinephrine and cortisol • Lipolysis: **Very active**; Increases free fatty acid levels in circulation • Insulin levels: High and insulin resistance • Ketone body levels in circulation – **LOW** • Increased ketone body utilization by peripheral tissues (due to hypermetabolism)	• Hormone sensitive lipase: Phosphorylated and active due to low levels of insulin • Lipolysis: Active; Increases free fatty acid levels in circulation • Insulin levels: Low • Energy for the Brain: Glucose + Ketone bodies • Ketone body levels in circulation - **High**

Flow phase: Protein Catabolism



Protein catabolism measured as urinary nitrogen (urea)

- Proportional to severity of illness
- **Negative nitrogen balance** during 'flow' phase
- In prolonged starvation, adaptive decrease in protein catabolism ('protein sparing' increases survival)
- Both critical illness and starvation are states of negative nitrogen balance, but protein depletion is more severe in critical illness

Flow phase: Dietary Protein Requirements

Disease process	Dietary Proteins (g/kg/day)
Basal (normal)	0.8-1.0
Postop (uncomplicated)	1.0-1.5
Sepsis	1.5-2.0
Multiple trauma (ventilator)	1.5-2.0
Major burns	2.0-3.0

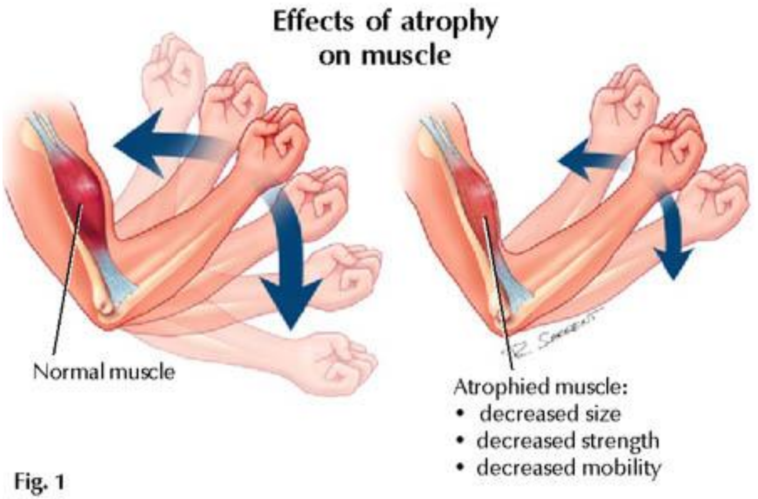


Fig. 1

Protein needs increase as severity of illness increases
Significant loss of body (tissue/muscle) protein

Protein metabolism in 'flow' phase

- Substantial **proteolysis** (catabolism) of muscle proteins (cortisol and insulin resistance; cytokines) → Cachexia
- Ubiquitin-proteasome system in muscle active (Somatic protein)
- **Reduced** protein synthesis and amino acid uptake by muscle
- Muscle proteolysis releases amino acids into circulation
- Amino acid catabolism in liver forms ammonia which forms **urea** (Increased urea formation)
- **Urine urea excretion** (gms/day) = extent of muscle proteolysis
- The intake of nitrogen (dietary protein) <<<<<**Output of nitrogen** (urinary nitrogen or urinary urea), **with a simultaneous breakdown of tissue protein**

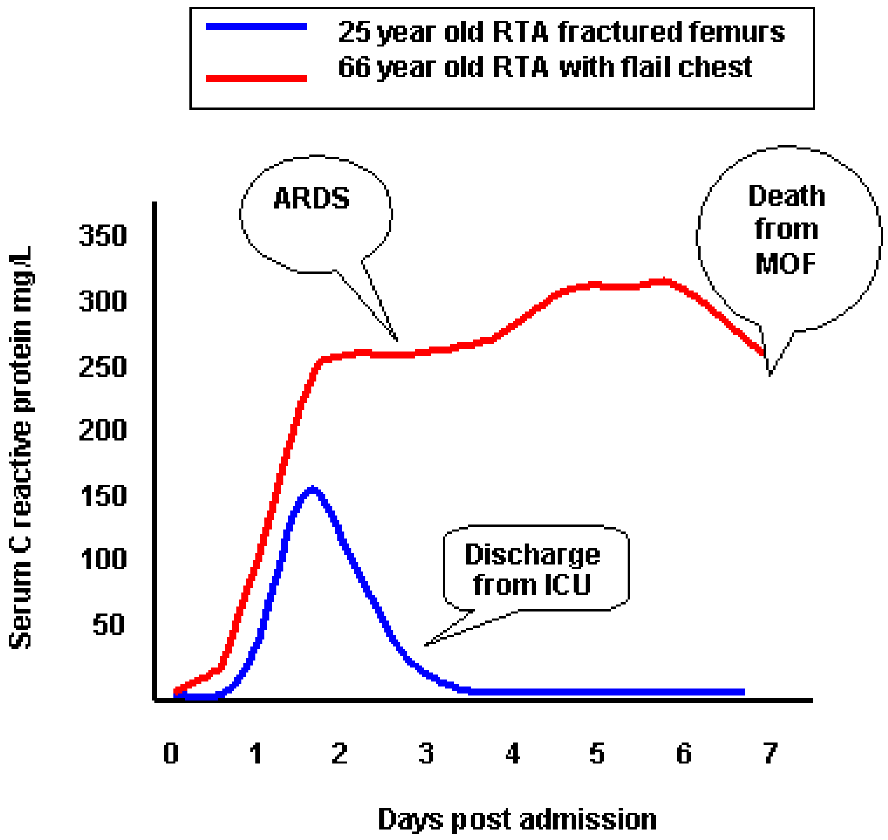
Uses of amino acids following muscle proteolysis

- Skeletal muscle (Somatic) major source of amino acids
  - Glutamine and alanine
  - Used for hepatic gluconeogenesis (C-skeleton) and amino group forms urea
  - Increased urea formation and increased urinary nitrogen loss
  - Maintenance of immune system (Immunoglobulin synthesis)
  - Acute-phase protein synthesis by liver
  - Nutritional Support and the Surgical Patient

Acute phase response by liver

- Cytokines stimulate liver to synthesize ‘**acute phase proteins**’
- Synthesized in increased amounts when there is inflammation
- **Not specific** for inflammatory process
- Positive acute phase proteins are **C-reactive protein, α_1-antitrypsin, ceruloplasmin, haptoglobin**
- Albumin is ‘negative’ acute phase protein (albumin synthesis decreases following an inflammation)

Acute phase response by liver



- Acute phase proteins (C-reactive protein, α_1-antitrypsin)
  - Monitor **progress** and **prognosis** of inflammation
  - Degree of rise of acute phase proteins proportional to severity of injury
  - Normalization of CRP levels indicate good response to therapy and good prognosis

Adverse effects of excessive protein catabolism

- Extensive visceral protein breakdown, **compromised** ability to adapt
- When 20-30% of body protein depleted, generally fatal
- Extensive **visceral protein depletion** results in
  - Impaired wound healing
  - Decreased immune response
  - Breakdown of gut-mucosal barrier (Visceral protein wasting)
  - Decreased mobility/ respiratory effort (respiratory muscle wasting) – risk of respiratory infections/ bed sores
  - Increased risk of infection and hypermetabolism (vicious cycle)
  - Multi-Organ failure

Comparison of protein metabolism

Critical illness (Flow phase)	Prolonged starvation
• **Muscle proteolysis**>>> Muscle protein synthesis • Very active Proteolysis (protein catabolism)→ **Muscle wasting** • **Negative nitrogen balance:** Marked and depends on severity • Urinary urea (Urinary nitrogen) excretion **elevated:** estimate degree of proteolysis	• Muscle proteolysis> Muscle protein synthesis • Proteolysis (protein catabolism) – active: Muscle wasting not as severe (Somatic) • Negative nitrogen balance (not as severe): Protein sparing effect of ketone bodies in prolonged starvation • **Urinary urea excretion reduced** (Protein sparing effect of ketone bodies)

Summary: Metabolic Response in 'Flow' phase of critical illness

- Involves most metabolic pathways
- Increased epinephrine, cortisol, cytokines and insulin (insulin resistance)
- Accelerated metabolic rate (hypermetabolic state)
- Negative nitrogen balance
- Muscle wasting due to excessive muscle proteolysis
- Acute phase protein synthesis in liver
- Increased urea excretion in urine

Amino acid/ Protein metabolism

- Increased adipose tissue lipolysis and serum free fatty acid levels
- Plasma ketone body levels are NOT increased

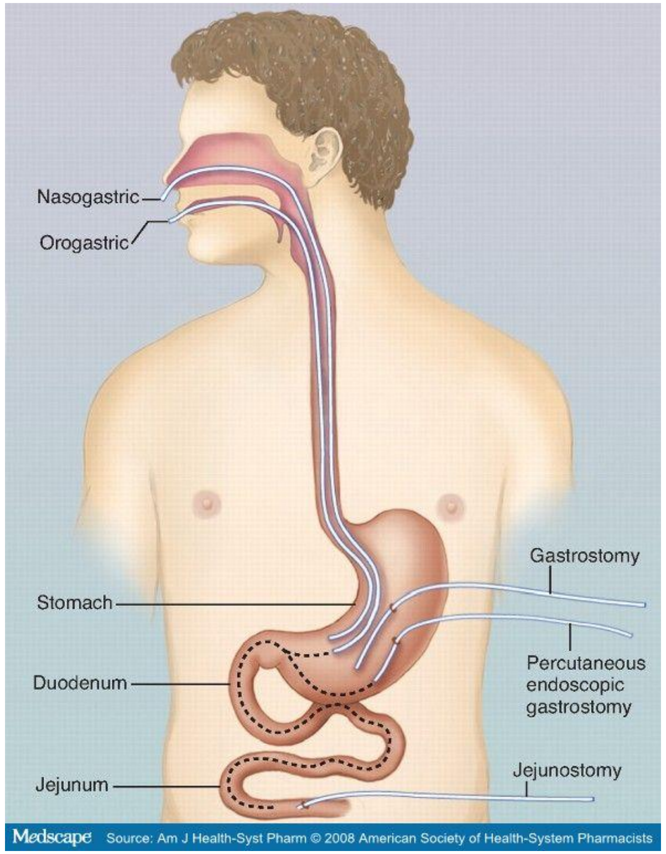
Lipid metabolism

- Insulin resistance
- Hyperglycemia
- Increased gluconeogenesis by liver

Carbohydrate metabolism

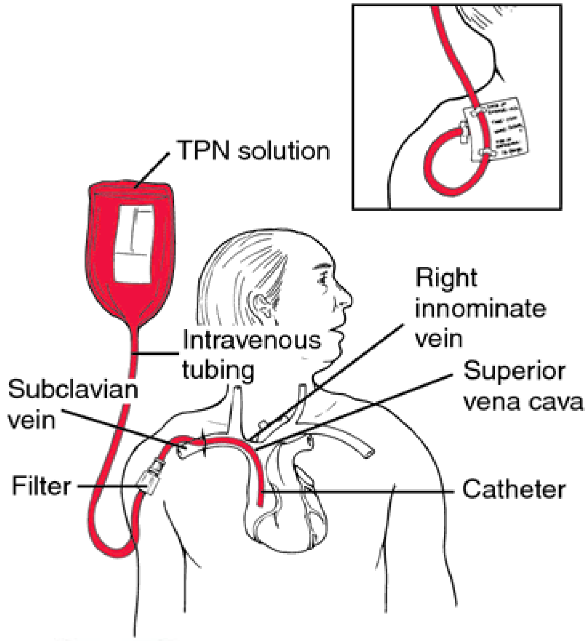
Enteral nutritional support (via GIT/intestine)

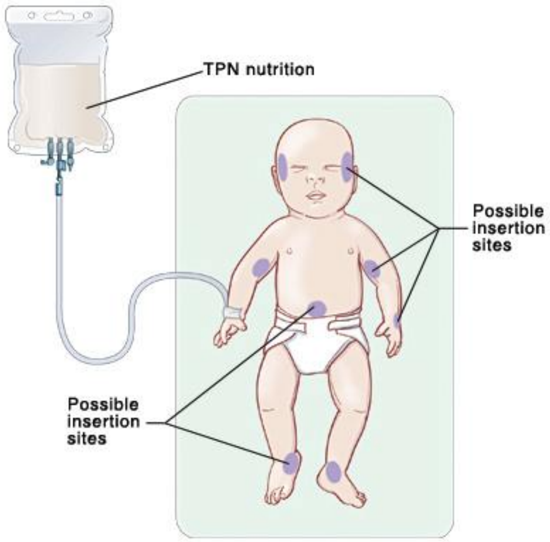
- Preserves intestinal mucosal integrity better than parenteral nutrition
  - Decreased mortality, decreased risk of bacterial translocation to mesenteric lymph nodes
  - Lower risk of sepsis (Gut an important source of bacteria)
  - Safer, convenient and less expensive
  - Preserves normal GI flora better
- Prerequisite for enteral nutrition: functional GI tract
- **Complication:** Aspiration of stomach contents



Parenteral (intravenous) nutrition

- Nutrients directly into systemic circulation (large vein; subclavian vein)
- Bypasses normal digestion and absorption in bowel
- **Sterile**, special liquid mixture given intravenously
- Used in patients who cannot eat or absorb enough nutrients to maintain good nutritional status
- **Life-saving** when used short term
- **Higher risk of infection and morbidity and mortality; Expensive**
- Recommendation: Begin enteral feeding ASAP





Anabolic phase (recovery)

- Positive nitrogen balance and build-up of tissue proteins
- Increase protein intake to allow tissue protein synthesis
- **Mobilization** and **activity** – facilitate muscle anabolism
- **Nitrogen intake** (2-2.5 g/kg body weight)>>> Nitrogen output in urine, and there is build-up of body protein (Anabolism)
- Rebuilding of adipose tissue stores
- Normalization of plasma glucose levels, normal insulin levels
- Tissues regain sensitivity to insulin
- Lasts for weeks-months following injury

Vitamins and minerals

- Vitamin C facilitate wound healing (prolyl hydroxylase for collagen)
  - Vitamin C - antioxidant
- Thiamine, niacin (Hypermetabolism increases requirements)
- Zinc improves wound healing, maintains immune function and improves appetite
- Copper (lysyl oxidase to form crosslinks in collagen) improves wound healing
- Omega-3 fatty acids improve inflammatory response (immunomodulators)